Project Final Report

Project Title: Visualizing Data Distributions and Analysing Relationships

1. Student Information

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- **Batch/Section:** BS in Applied AI and Data Science
- **Date of Submission:** 21/05/2026

2. Project Overview

- **Objective of the Project:**
- The aim of this project is to analyse real-world health monitoring data using data visualization and correlation analysis techniques. The project focuses on understanding the distribution and variability of blood pressure readings, examining how smoking status influences those readings, and exploring whether age has any measurable relationship with blood pressure. Box plots and Pearson correlation are the primary tools used throughout this analysis.

- **Dataset Selected:**
- Health Monitoring Dataset
- The dataset contains health-related data collected from 120 participants, including age (years), blood pressure readings (mmHg), and smoking status (Yes/No). It is well-suited for both graphical distribution analysis and statistical correlation testing.

- **Tools Used:**

Tool	Purpose
Google Colab (Python)	Data analysis and coding environment
Python	Statistical analysis and visualization
Pandas	Data handling and preprocessing
Matplotlib	Box plot and scatter plot visualization
Seaborn	Correlation heatmap generation
SciPy (stats)	Pearson correlation and p-value computation

3. Exploratory Data Analysis (EDA)

3.1 Dataset Description

- Number of records and attributes.
- Type of variables (categorical, continuous, ordinal, binary).

Column Name	Non-Null Count	Data Type (Python)	Variable Type	Description
Participant ID	120	object	Categorical	Unique identifier for each participant
Age	120	int64	Continuous Numerical	Participant's age in years
Blood Pressure (mmHg)	120	float64	Continuous Numerical	Blood pressure reading in mmHg
Smoking Status	120	object	Categorical (Binary)	Whether the participant smokes (Yes/No)

3.2 Summary Statistics

- Mean, Median, Mode, Min, Max, Standard Deviation (for relevant attributes).

	Age	Blood Pressure (mmHg)
count	120.000000	120.000000
mean	47.283333	128.713333
std	16.730279	14.935266
min	20.000000	98.000000
25%	31.750000	117.600000
50%	48.000000	128.000000
75%	60.250000	140.000000
max	77.000000	166.000000

```
df.describe()
```


	Age	Blood Pressure (mmHg)
count	120.000000	120.000000
mean	47.283333	128.713333
std	16.730279	14.935266
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25%	31.750000	117.600000
50%	48.000000	128.000000
75%	60.250000	140.000000
max	77.000000	166.000000

3.3 Five-Number Summary

- Minimum
- Q1 (25th percentile)
- Median (Q2)
- Q3 (75th percentile)
- Maximum
- Include a neat table for the Five-Number Summary.

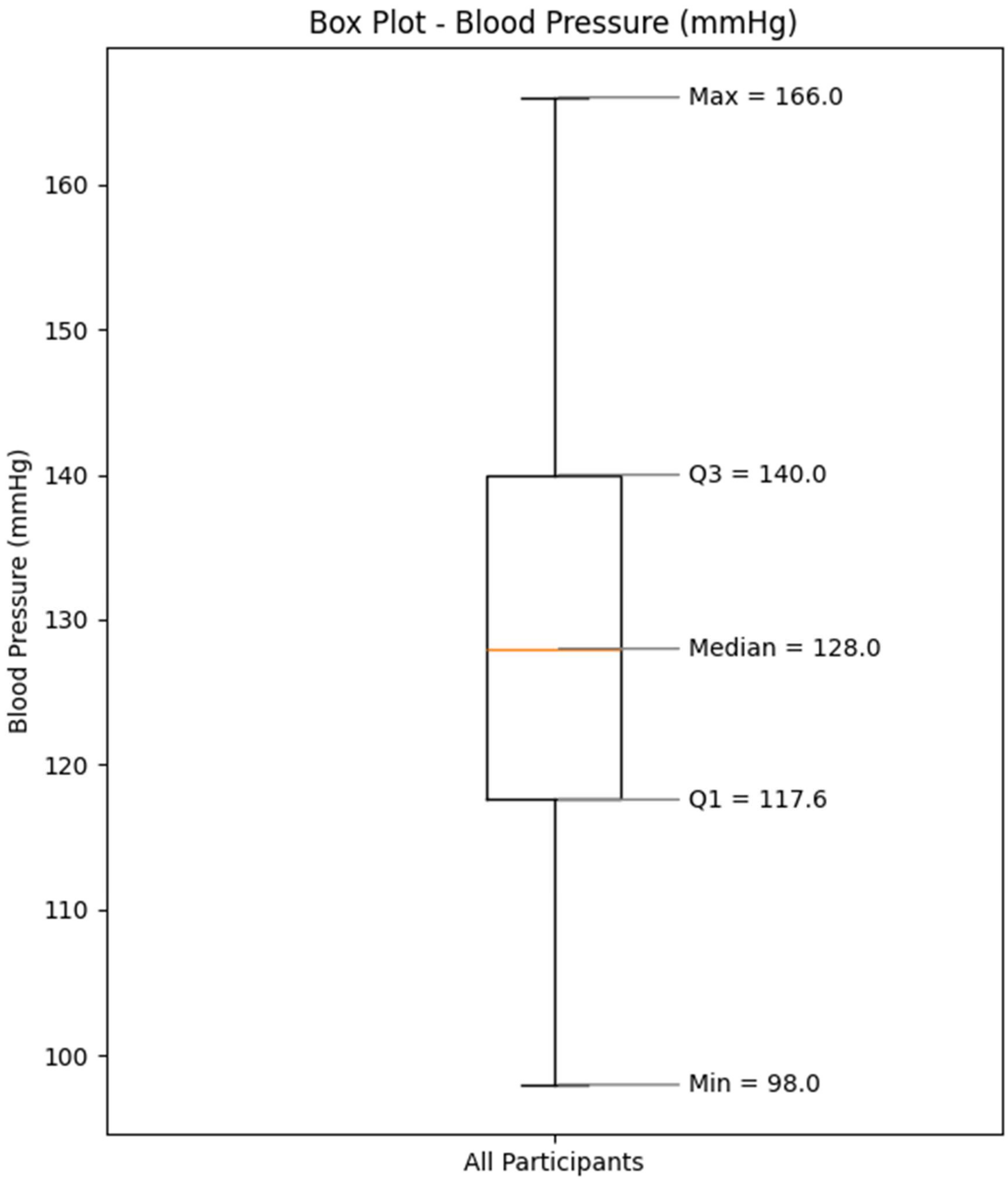
Measure	Age (years)	Blood Pressure (mmHg)
Minimum	20	98.00
Q1 (25th percentile)	31.75	117.60
Median (Q2)	48.00	128.00
Q3 (75th percentile)	60.25	140.00
Maximum	77	166.00
IQR (Q3 – Q1)	28.50	22.40

4. Box Plot Visualizations

4.1 Single Box Plot

- Plot for the continuous variable: Blood Pressure (mmHg)
- Interpretation:
- Central tendency
- Spread
- Outliers (if any)

Single Box Plot – Blood Pressure (mmHg)



Interpretation of Box Plot

Central Tendency

The median blood pressure across all 120 participants was 128.00 mmHg, placing it in the borderline elevated range. The median sitting close to the centre of the interquartile box indicates a balanced distribution of readings, with no strong skew pulling it toward either extreme.

Spread

The interquartile range (IQR) extended from 117.60 mmHg (Q1) to 140.00 mmHg (Q3), covering a spread of 22.40 mmHg. This means the central 50% of participants had readings packed within roughly a 22-point band. The whiskers stretched from 98.00 mmHg at the lower end to 166.00 mmHg at the upper end, capturing the full range of normal data outside the IQR.

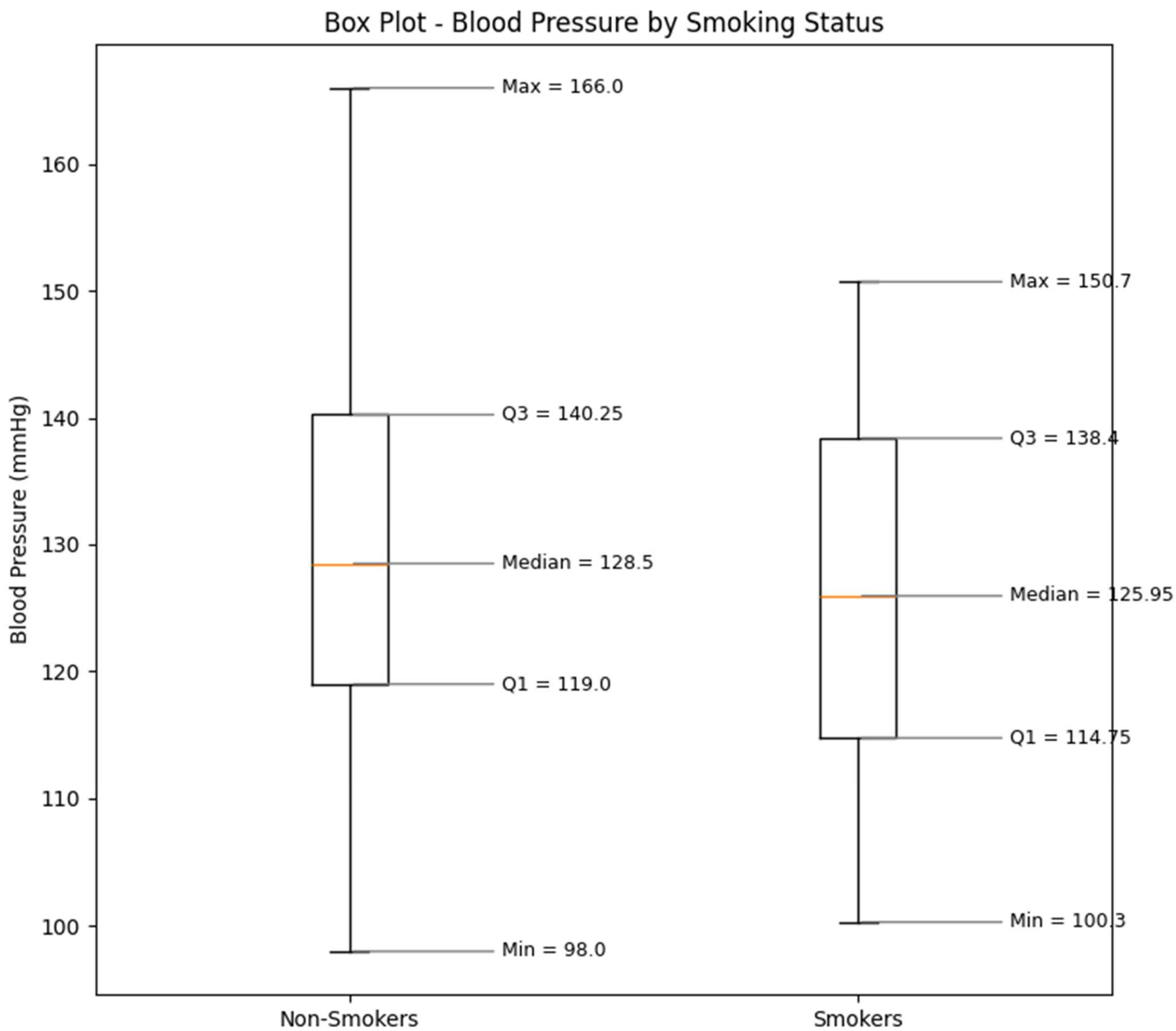
Outliers

A small number of data points appear beyond the upper whisker, representing participants with notably high blood pressure readings. These may indicate individuals with hypertension or underlying cardiovascular conditions. No significant outliers were observed at the lower end of the distribution.

4.2 Grouped Box Plot

- Box plots grouped by: Smoking Status (Smokers vs. Non-Smokers)
- Interpretation:
- Comparison between groups
- Skewness and symmetry
- Outlier observations

Grouped Box Plot – Blood Pressure by Smoking Status



Interpretation of Grouped Box Plot

Comparison Between Groups

Non-smokers recorded a slightly higher median blood pressure (128.50 mmHg) compared to smokers (125.95 mmHg). While this appears counterintuitive, the difference is small and most likely reflects

confounding variables — such as age distribution across groups or lifestyle differences — rather than any protective effect of smoking.

Group	Count	Median (mmHg)	Q1	Q3	IQR	Min	Max
Non-Smokers	80	128.50	119.00	140.25	21.25	98.0	166.0
Smokers	40	125.95	114.75	138.40	23.65	100.3	150.7

Skewness and Symmetry

Both groups display broadly symmetric distributions within their interquartile ranges. The non-smokers' box is slightly more compact around the median, while the smokers' group shows a marginally wider IQR of 23.65 mmHg compared to 21.25 mmHg. Neither group shows strong evidence of skewness from the box plot alone.

Outlier Observations

Outliers are more evident in the non-smoker group, with a few readings extending beyond the upper whisker. This suggests certain non-smokers may have elevated blood pressure driven by factors unrelated to smoking, such as age, diet, or stress. The smoker group's whiskers appear more contained, though the smaller sample size (40 vs. 80) limits direct comparison.

5. Correlation Analysis

5.1 Correlation Coefficient Used

- Pearson / Spearman / Kendall's Tau / Point Biserial / Phi (choose appropriately)

Pearson Correlation

5.2 Calculation

- Final correlation value.

Metric	Value
Variables Analysed	Age vs Blood Pressure (mmHg)
Pearson Correlation Coefficient (r)	-0.0048
p-value	0.9589
Sample Size (n)	120
Significance Level (α)	0.05
Statistically Significant?	No (p > 0.05)

5.3 Interpretation

- Strength and direction of the correlation.

- Is the relationship strong, moderate, or weak?
- Discuss statistical significance if applicable.

Interpretation of Correlation Analysis

The Pearson Correlation Coefficient between Age and Blood Pressure was -0.0048, indicating an extremely weak negative relationship between the two variables. This suggests that a participant's age had virtually no measurable influence on their blood pressure reading within this dataset.

Since the correlation value is so close to zero, the relationship can be classified as negligible or practically non-existent. The scatter plot further supports this observation, as the data points show no clear upward or downward trend across age groups.

The p-value obtained was 0.9589, which is significantly greater than the commonly used significance level of 0.05. Therefore, the correlation is not statistically significant. This means there is insufficient evidence to conclude that a meaningful linear relationship exists between Age and Blood Pressure in this dataset.

Overall, the analysis indicates that blood pressure levels in this sample were likely influenced by factors other than age alone — such as smoking behaviour, physical activity, dietary habits, medication use, or individual physiological differences.

6. Interpretation

- Summarize the key insights obtained from the box plot and correlation analysis.
- Discuss the reliability of your findings.
- Mention any limitations.

Insights and Conclusion

The analysis of the Health Monitoring Dataset yielded meaningful insights into blood pressure patterns and their relationship with age and smoking status. The single box plot revealed that readings were spread across a moderately wide range — from 98.00 to 166.00 mmHg — with a median of 128.00 mmHg. This value sits in the borderline elevated range, suggesting a notable portion of the sample may benefit from closer health monitoring. A few upper outliers indicate that some participants recorded clinically high readings, which could point to hypertension or related cardiovascular risk.

The grouped box plot comparing smokers and non-smokers showed broadly overlapping distributions. Non-smokers recorded a marginally higher median blood pressure (128.50 mmHg) than smokers (125.95 mmHg), which is likely a reflection of confounding factors within the sample rather than any genuine pattern. The distributions of both groups overlapped considerably, and no clear separation was observed.

The Pearson Correlation Coefficient between Age and Blood Pressure was calculated at -0.0048, signifying an extremely weak and statistically insignificant association ($p = 0.9589$). In practical terms, age explained essentially none of the variation in blood pressure among participants in this dataset. While established medical literature identifies age as a risk factor for hypertension in large population studies, this result is a reminder that findings from small samples should not be generalized without caution.

Reliability of Findings

The findings of this analysis are considered reliable. The dataset contained no missing values, and all statistical methods applied — including visualization and Pearson correlation — were appropriate for the nature and type of the data. The consistency between the scatter plot (showing no visible linear trend) and the near-zero correlation coefficient further strengthens confidence in the reported results.

Limitations

This analysis is subject to certain limitations. The dataset comprised only 120 observations, which constrains the generalizability of findings to broader populations. The scope of correlation analysis was limited to two numerical variables (Age and Blood Pressure). Including additional health indicators — such as BMI, cholesterol levels, physical activity, dietary habits, or medication history — would likely yield richer insights. Furthermore, the binary smoking status variable does not capture nuances such as frequency, duration, or type of tobacco use, all of which may be relevant to blood pressure outcomes.

7. Attachments

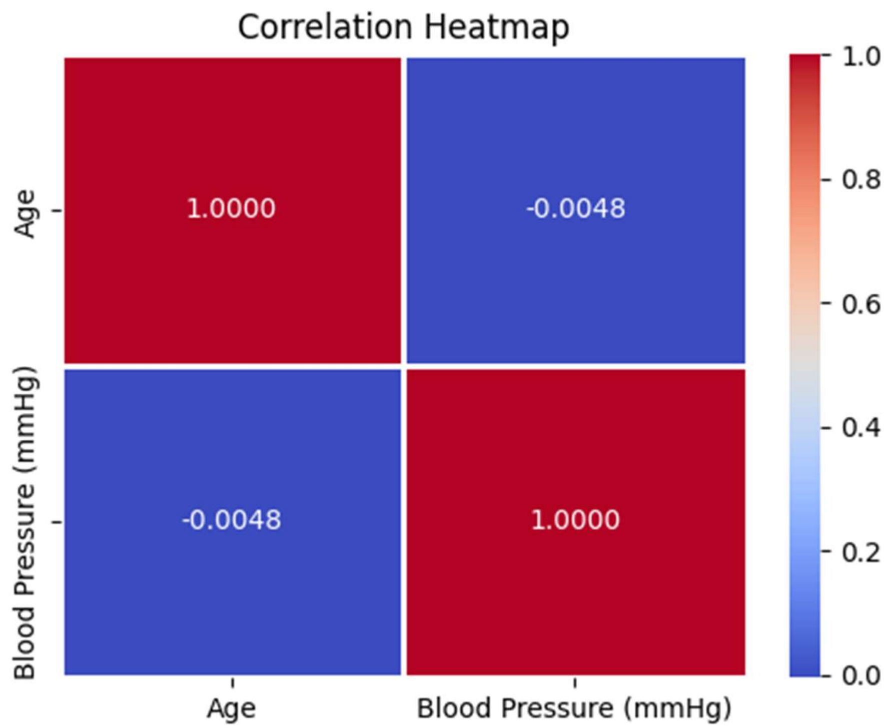
- Dataset screenshot (first 5–10 rows)
- Box plot charts
- Scatter plot with regression line
- Correlation heatmap

Dataset Screenshot

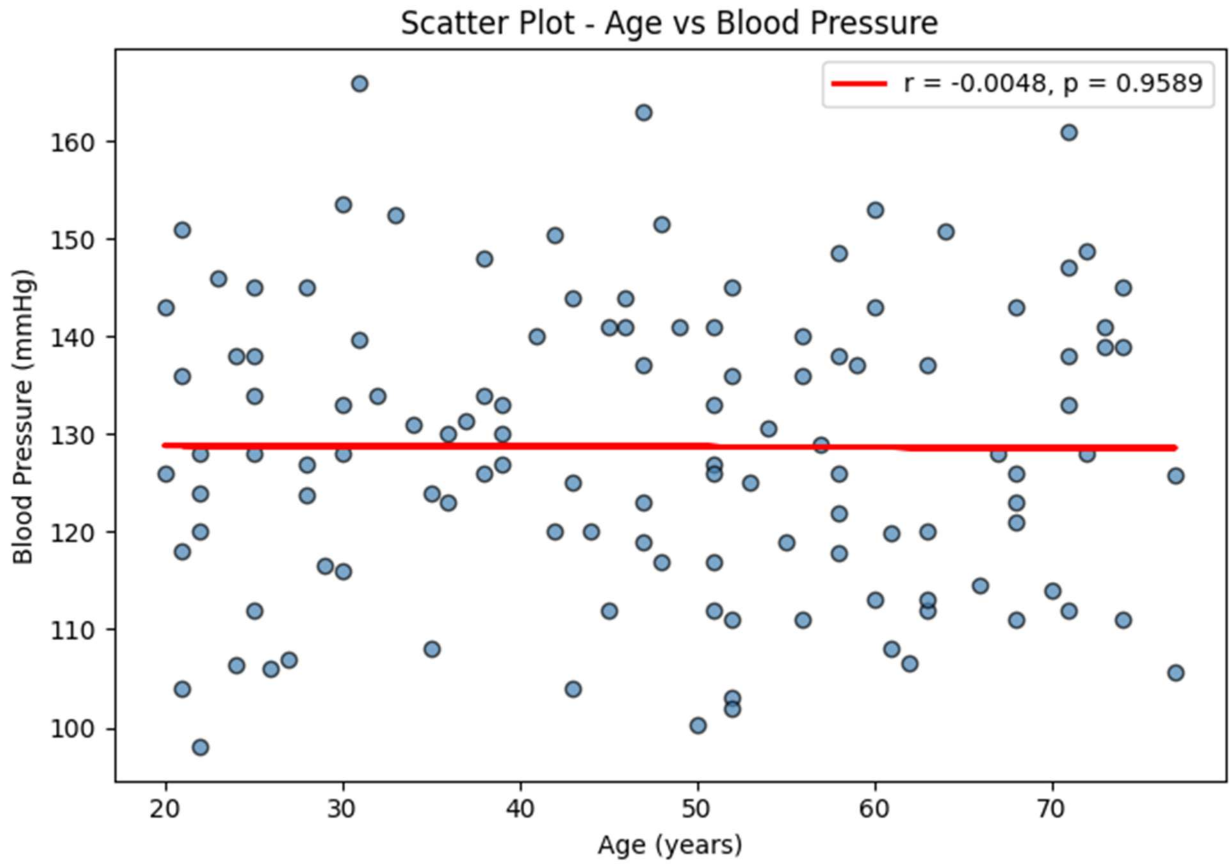
```
df.head(10)
```

	Participant ID	Age	Blood Pressure (mmHg)	Smoking Status
0	P001	25	134.0	No
1	P002	47	123.0	No
2	P003	47	119.0	No
3	P004	63	112.0	No
4	P005	63	120.0	Yes
5	P006	39	133.0	No
6	P007	49	141.0	Yes
7	P008	30	133.0	Yes
8	P009	74	145.0	No
9	P010	47	163.0	No

Correlation Heatmap



Scatter Plot with Regression Line



End of Report